

EE5507 Project 1 Report

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No.	Specification	Value	Achieved	Unit / Condition
1	DC Gain	> 70	78	dB
2	GBW	> 15	30	MHz ($C_{Load} = 1$ pF)
3	Phase Margin	> 65	67	Degree
4	Input Offset Voltage	< 5	1.43	mV
5	Input Common Mode Range	$0.1 - 1.2$	$0 - 1.2$	V
6	Output Common Mode Range	$0.3 - 1$	$0.21 - 1$	V

Table 1: OTA Design Specifications and Achieved Performance

1 Topology Selection

From the specification, the hardest to achieve (based on lecture content) is the 0.1 V to 1.2 V input common mode swing. The closest achievable topology would be a folded cascode op-amp, which allows the input CM to swing above or below one supply rail (if the cascode is biased appropriately), but the swing in the other direction is limited by the V_{GS} of the input transistor and V_{OD} of the tail current source. Since $V_{GS} > V_{TH} \approx 0.4$ V, reaching down to 0.1 V is not achievable.

To achieve rail-to-rail input CM swing, a complementary input pair is used, where one pair takes over when the other enters the linear region. However, when both pairs are simultaneously active, their g_m combines, causing a theoretical variation in overall g_m by a factor of 2. [1] explores different methods to reduce this g_m variation, and the minimum current selection approach by [2] was selected for its simplicity, as it requires no extra biasing voltages. A possible disadvantage of this approach is the current draw, since many branches are required to carry the same current as the differential pair tail current.

Figure 1 shows the implementation from [2]. From Tutorial 3, a 2-stage non-cascode op-amp achieved a gain of around 40 dB. Without drastically changing sizing or current, a single-stage cascode should also yield around 40 dB, since the $g_m r_o g_m r_o$ product is roughly the same. Therefore, on top of using Figure 1, a cascoded common-source second stage is added.

2 Achieved Performance

Figure 21 and Figure 22 show the sizing and DC operating points of the amplifier, with stages labelled to ease discussion henceforth (Note: these are high resolution, should have no issue zooming in). Figure 2 and Figure 3 show the sizing and DC operating points of the bias generator. Figure 4 and Figure 5 show the sizing and DC operating points of the minimum current selector. Figure 6 shows the top-level schematic measuring the open-loop and closed-loop response. Figure 7 shows the top-level schematic measuring the input offset voltage. The same schematic with DC operating

points, Figure 8, shows the total current draw to be 2.8 mA.

2.1 DC Gain & GBW

The differential mode open-loop gain under different input CM levels is plotted against frequency in Figure 9. The variation in differential mode open-loop gain against input CM level is plotted in Figure 10, showing a DC gain of at least 78 dB. The open-loop unity gain frequency (GBW) also varies with input level. As shown in Figure 9, the GBW is at least 30 MHz.

Across the input CM range, the DC gain varies within ± 0.65 dB $\approx \pm 10\%$, which is an improvement from the theoretical factor of 2 variation if there is no circuit to flatten the overall g_m . [2] obtained a $g_{m,\text{total}}$ variation of 7%. Similar to [2], the PMOS current source has a cascode while the NMOS current source does not. This makes the minimum current selection circuit operate more ideally when replicating PMOS current, i.e., when $I_{in2} > I_{in1}$, which occurs when the input CM is low and the input differential pair is predominantly done by PMOS. This may explain the asymmetrical behaviour in Figure 10 as the input CM is swept from 0 to 1.2 V.

To compensate, the NMOS length is set to 1 μm , which is longer than the PMOS length of 0.5 μm . A cascode was not added to the NMOS current source since it would require additional voltage bias generation, as reusing BN1 would force it into the linear region due to the next stage (STAGE 1C).

2.2 Phase Margin

Figure 11 shows the differential mode loop gain. With lead-lag compensation applied, the unity loop gain frequency is 19 MHz and the phase margin is 67° . Figure 12 shows the common mode loop gain, where it is far below 0 dB. Step response to differential mode input is shown in Figure 13, Figure 14, and Figure 15. Step response to common mode input is shown in Figure 16, Figure 17, and Figure 18.

2.3 Input Offset Voltage

Figure 19 shows the offset voltage plot with a Monte Carlo simulation of 500 samples, ranging from -3.8 mV to 4.3 mV. Taking the larger value of 4.3 mV, the input offset voltage is calculated:

$$\begin{aligned}\pm 3\sigma &= \pm 4.3 \text{ mV} \\ \sigma &= \frac{4.3}{3} \text{ mV} \\ &= 1.43 \text{ mV} \\ \sigma &= V_{\text{offset}} = \pm 1.43 \text{ mV}\end{aligned}$$

2.4 Input CM Range

From left half side STAGE 1A and STAGE 1B in Figure 22.

PMOS Input Differential Pair:

$$\begin{aligned}
V_{ic,max} &= V_{DD} - |V_{OD,PM47}| - |V_{GS,PM32}| \\
&= 1.2 - 0.073 - (0.991 - 0.6) \\
&= 0.736 \text{ V} \\
V_{ic,min} &= V_{BN1} - V_{GS,NM23} + |V_{OD,PM32}| - |V_{GS,PM32}| \\
&= 0.590 - (0.590 - 0.111) + 0.057 - (0.991 - 0.6) \\
&= -0.223 \text{ V}
\end{aligned}$$

NMOS Input Differential Pair:

$$\begin{aligned}
V_{ic,max} &= V_{BP2} + |V_{GS,PM10}| - V_{OD,NM21} + V_{GS,NM21} \\
&= 0.664 + (1.099 - 0.664) - 0.054 + (0.6 - 0.204) \\
&= 1.441 \text{ V} \\
V_{ic,min} &= V_{GS,NM21} + V_{OD,NM50} \\
&= (0.6 - 0.204) + 0.064 \\
&= 0.460 \text{ V}
\end{aligned}$$

Input CM range: -0.223 V to $1.441 \text{ V} \Rightarrow 0 \text{ V}$ to 1.2 V

2.5 Output CM Range

For the output, the CM voltage is held roughly at half the supply rail by the CMFB as shown in [Figure 20](#). But the maximum and minimum voltage swing can still be calculated. From the left half side STAGE 2 in [Figure 22](#):

$$\begin{aligned}
V_{oc,max} &= V_{DD} - |V_{DS,PM25}| - |V_{OD,PM33}| \\
&= 1.2 - 0.110 - 0.084 \\
&= 1.006 \text{ V} \\
V_{oc,min} &= V_{DS,NM37} + V_{OD,NM36} \\
&= 0.113 + 0.096 \\
&= 0.209 \text{ V}
\end{aligned}$$

Output CM range: 0.2 V to 1.0 V .

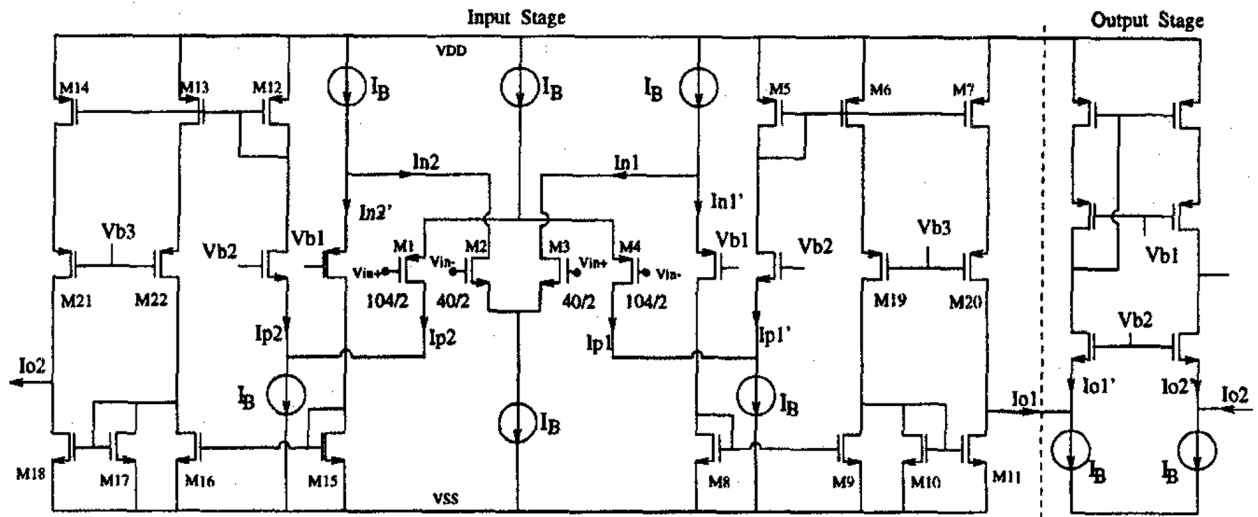


Figure 1: Schematic from [2]

Bias Generation

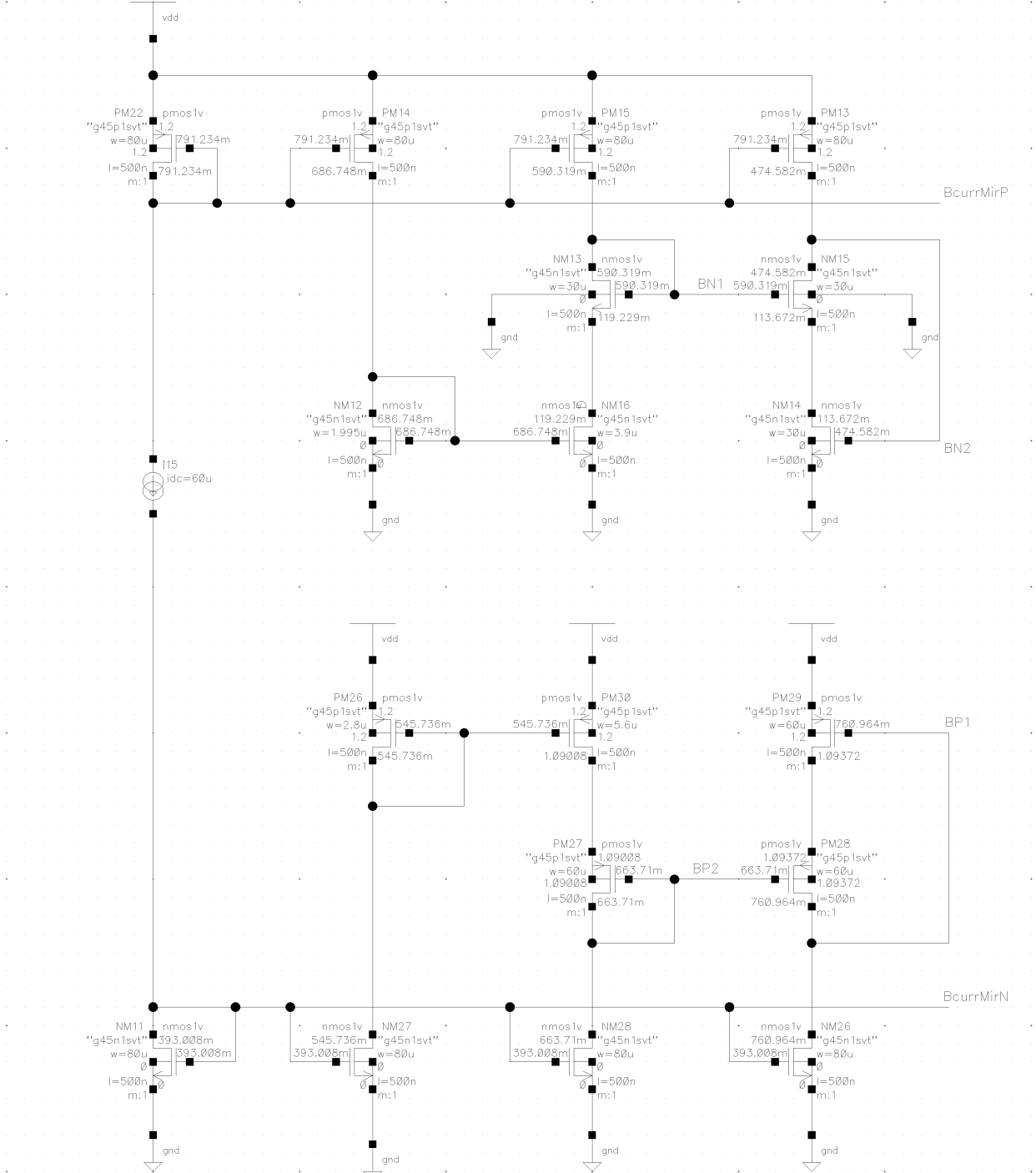


Figure 2: Final Schematic of Bias Generation (Sizing)

Bias Generation

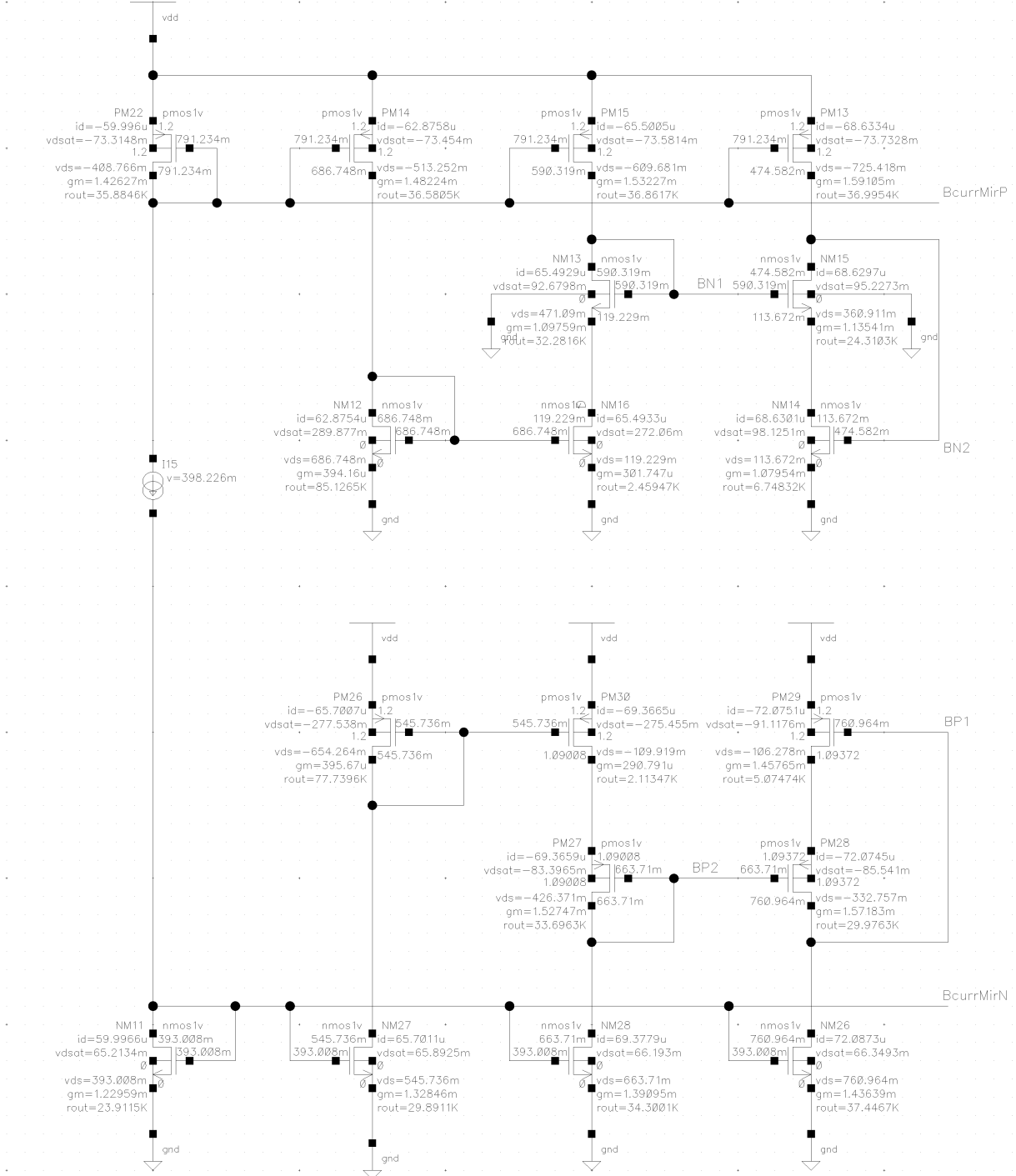


Figure 3: Final Schematic of Bias Generation (DC Operating Point at Input CM = 0.6 V)

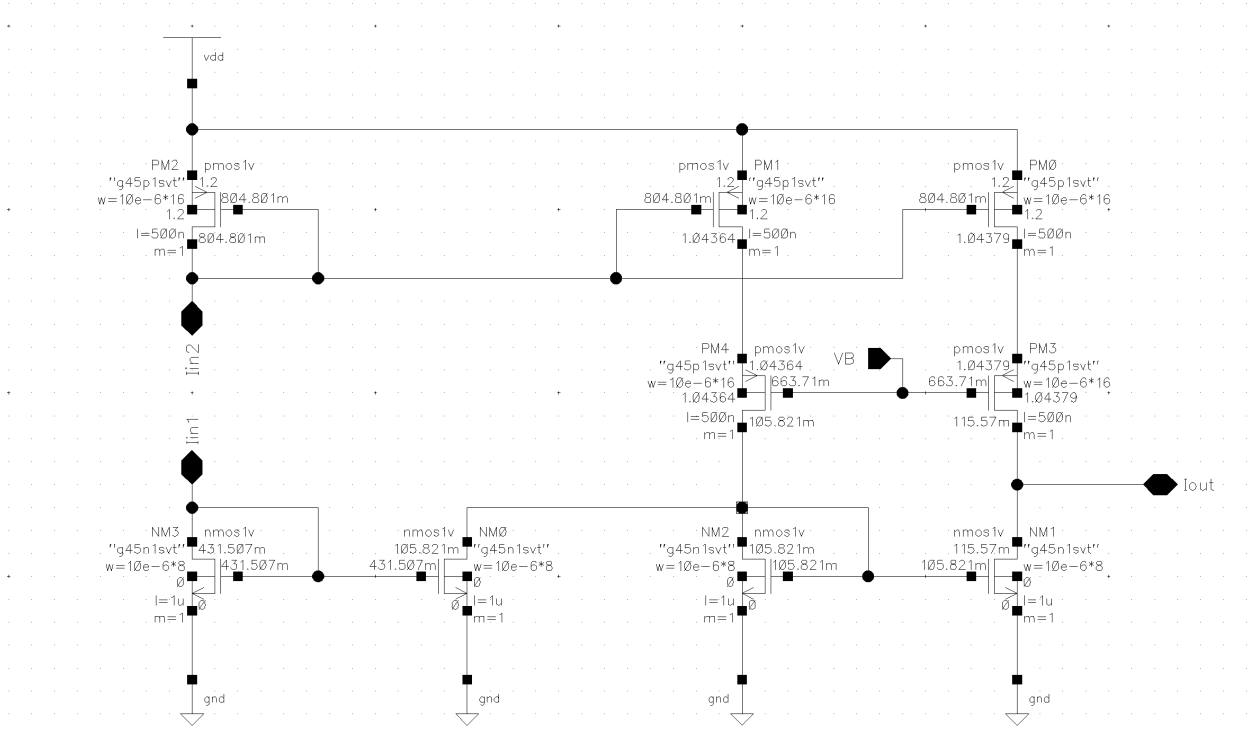


Figure 4: Final Schematic of Minimum Current Selector (Sizing)

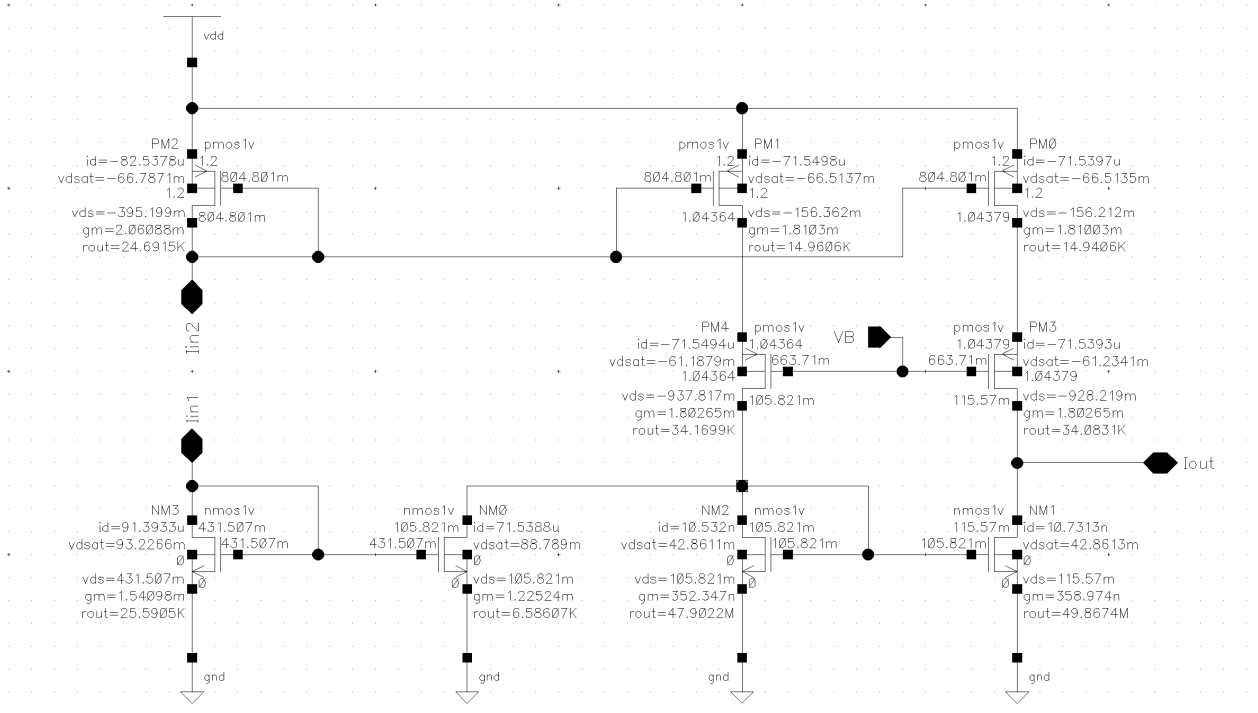


Figure 5: Final Schematic of Minimum Current Selector (DC Operating Point at Input CM = 0.6 V)

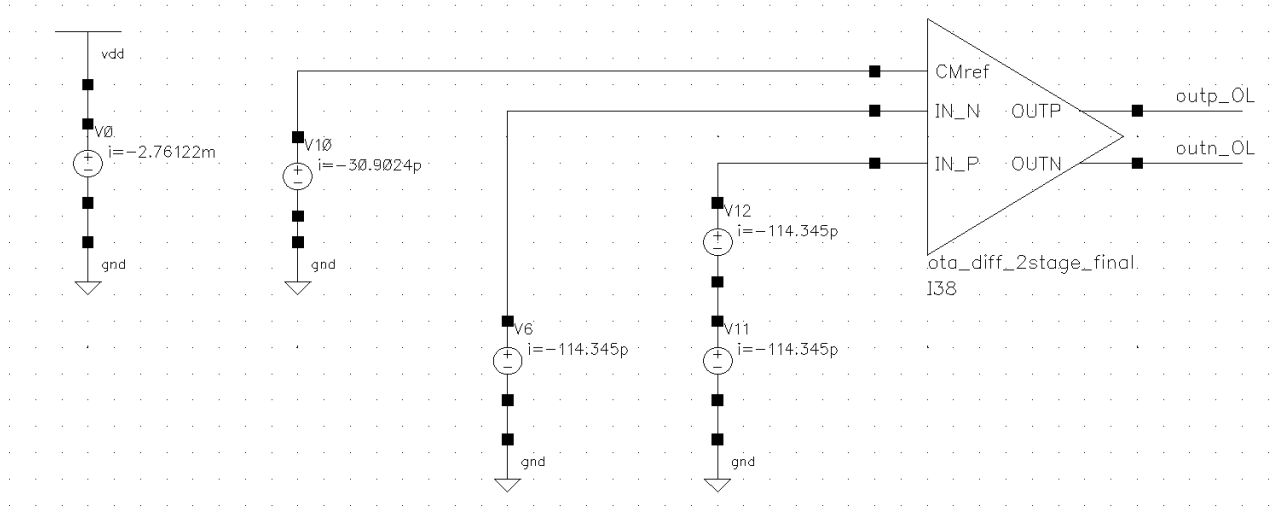


Figure 8: Top Level Schematic for Simulating Input Offset Voltage Showing Total Current Draw of OTA

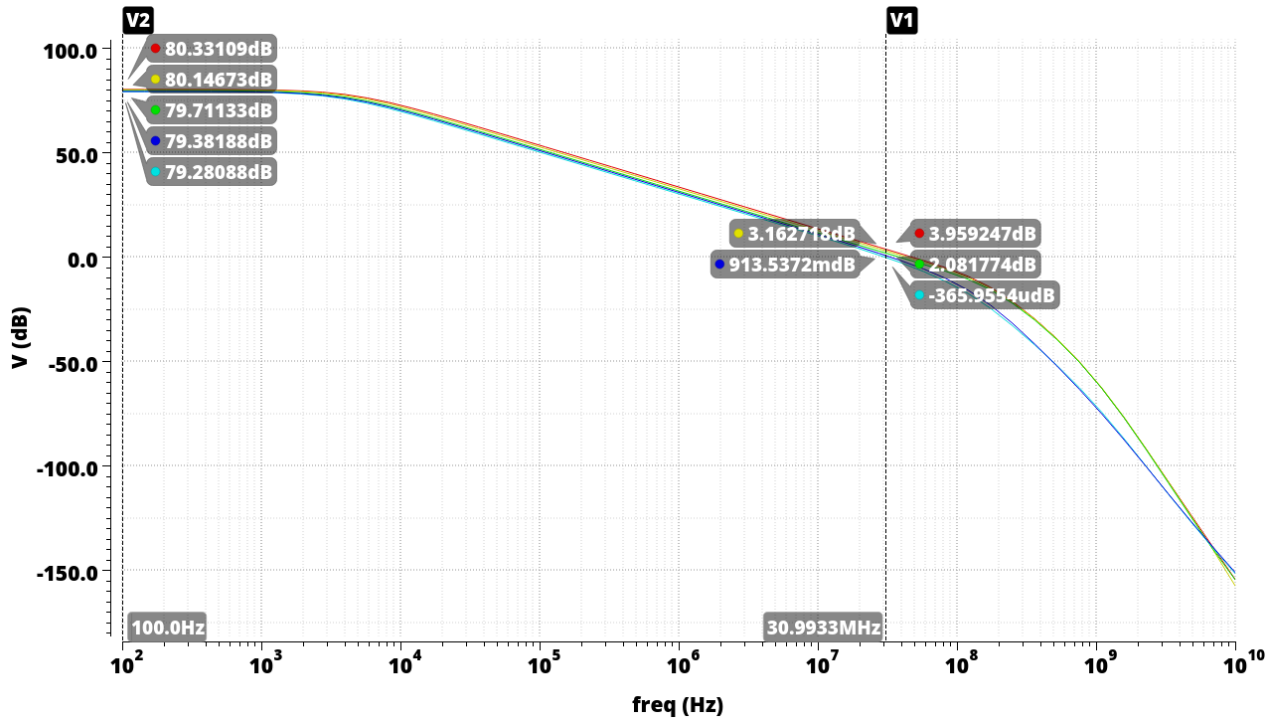


Figure 9: Differential Mode Open Loop Gain with Input CM = 0, 0.3, 0.6, 0.9 and 1.2 V. Corresponding to red, yellow, green, cyan and blue trace

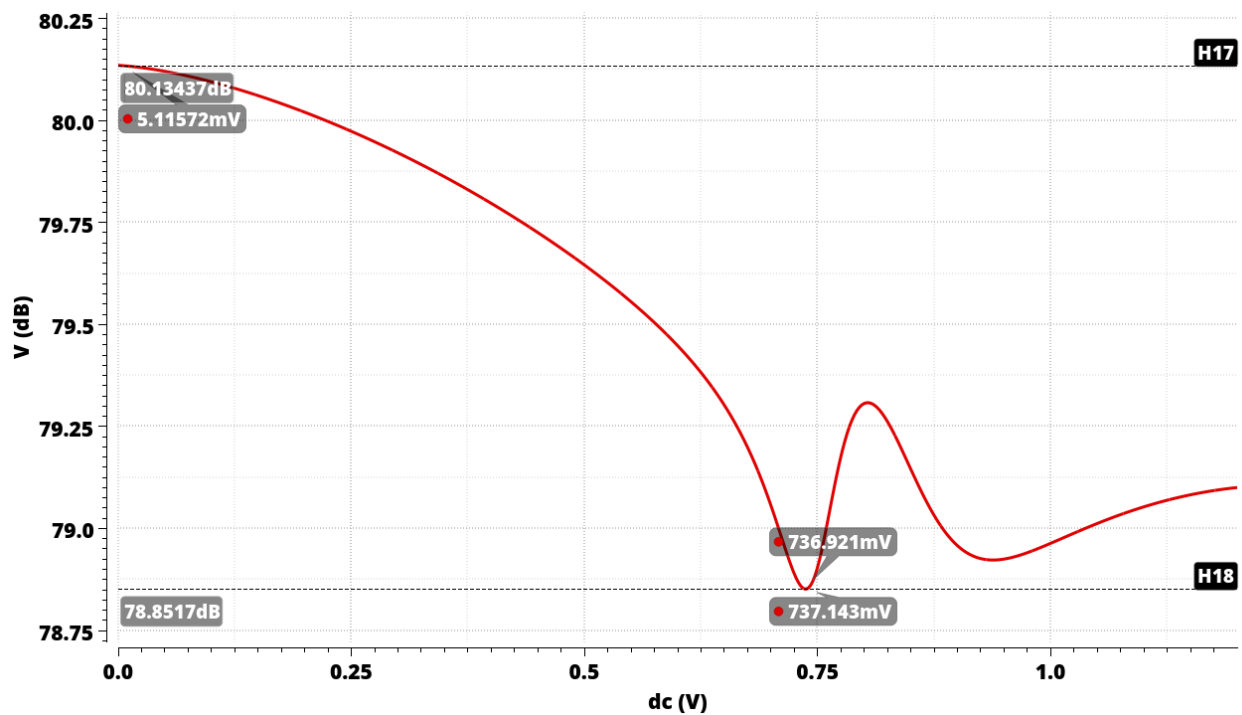


Figure 10: Differential Mode Open Loop Gain at 1 kHz vs Input CM

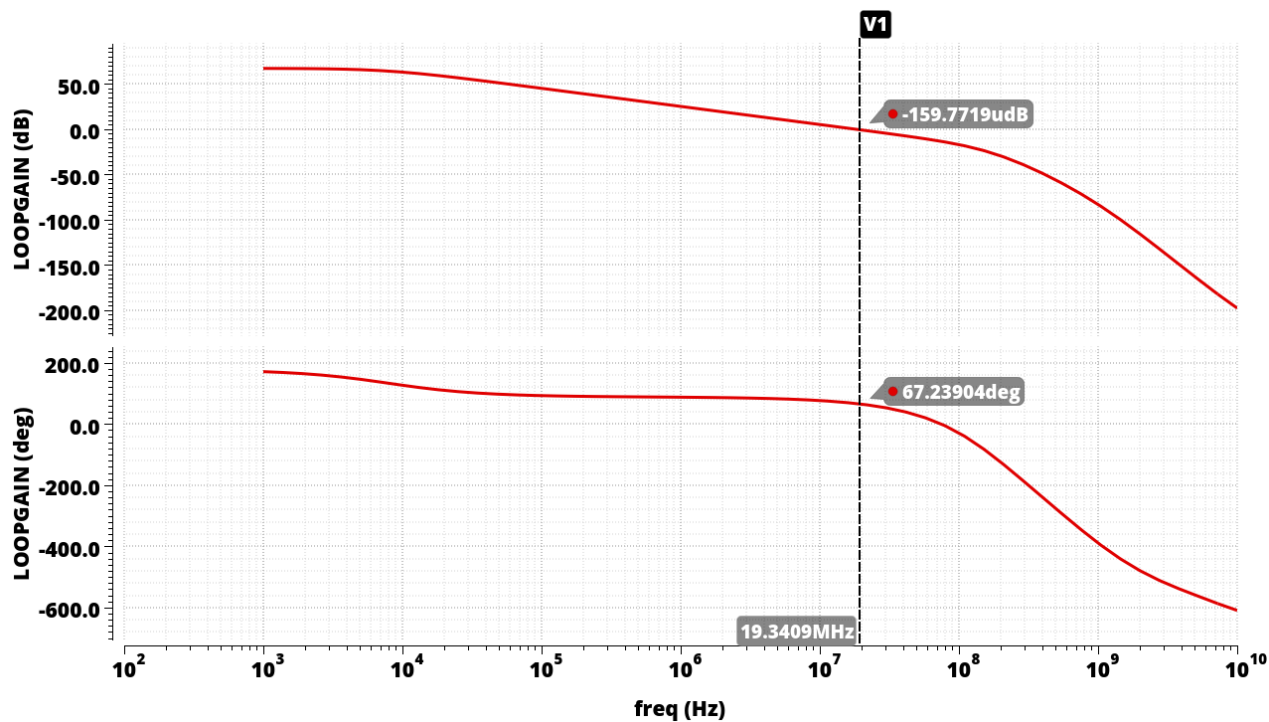


Figure 11: Loop Gain for Differential Mode

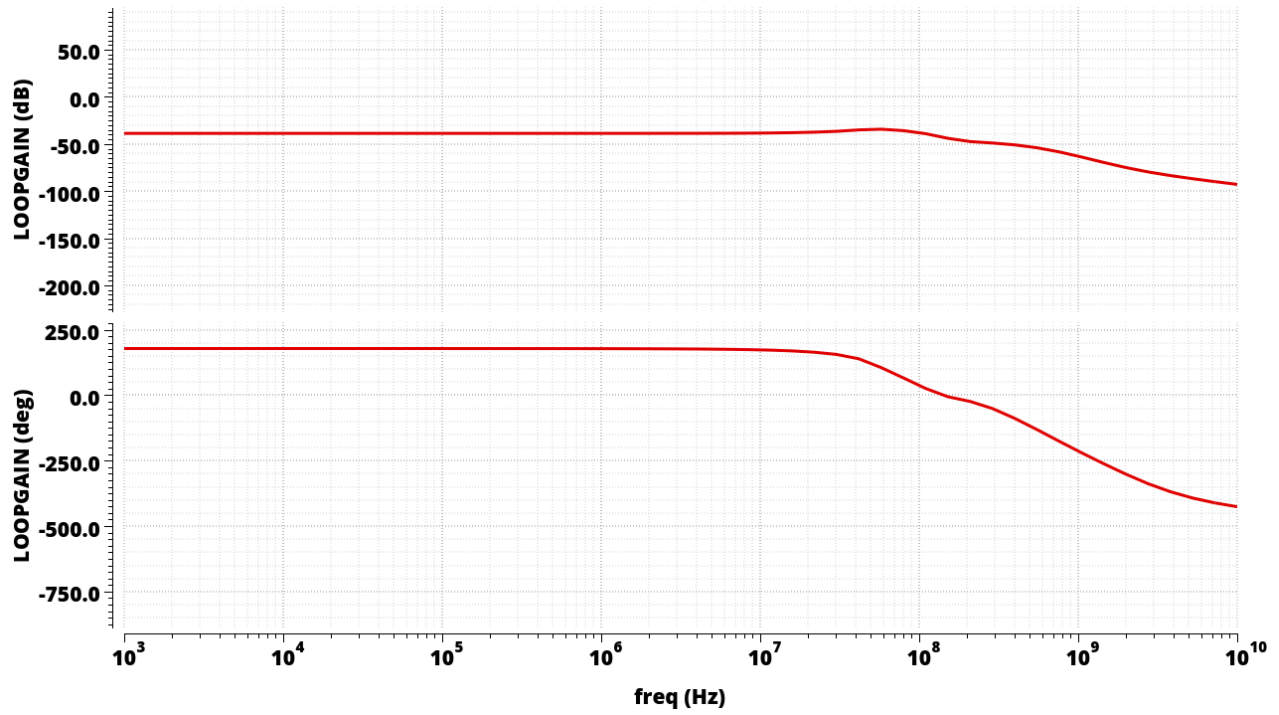


Figure 12: Loop Gain for Common Mode

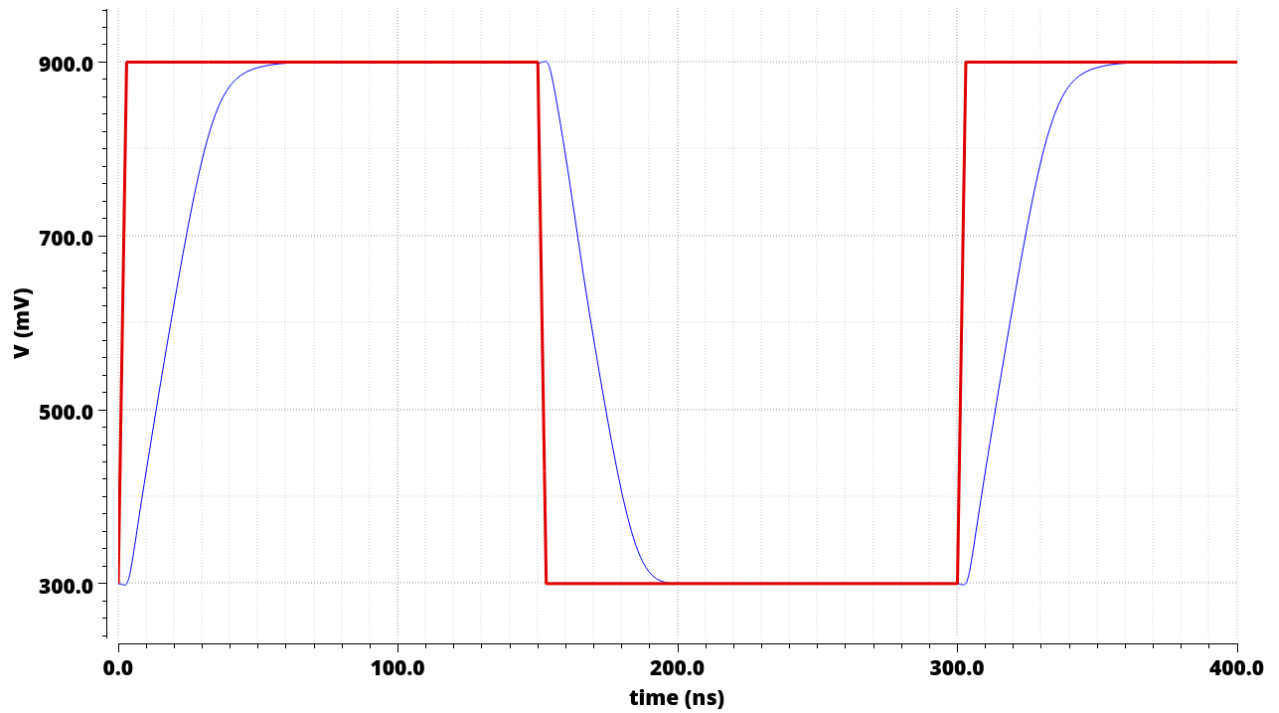


Figure 13: Step Response for Differential Mode

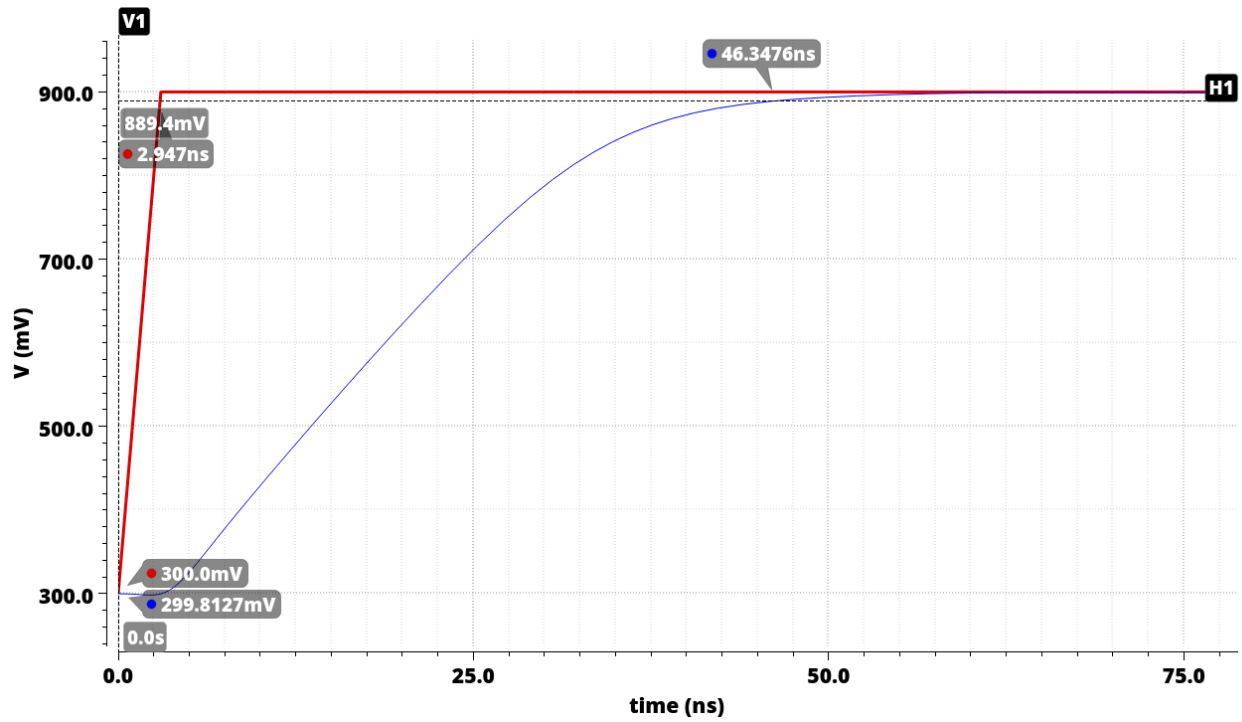


Figure 14: Step Response for Differential Mode (zoomed to rising edge)

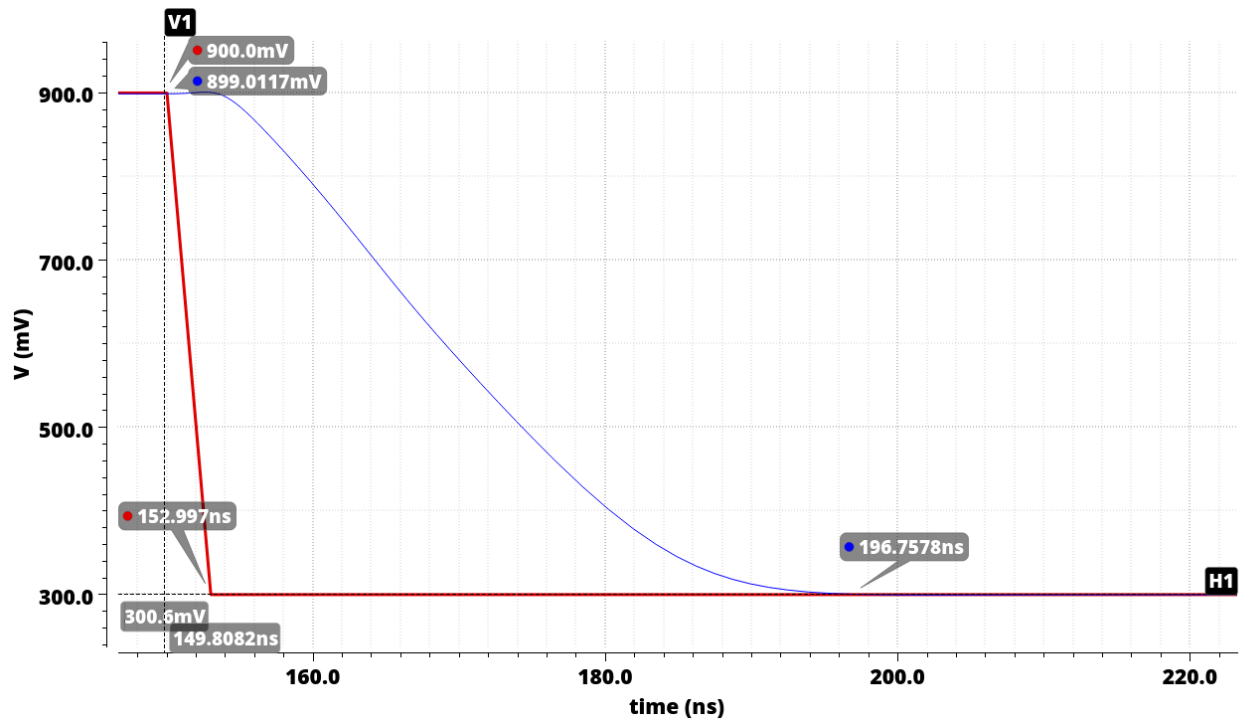


Figure 15: Step Response for Differential Mode (zoomed to falling edge)

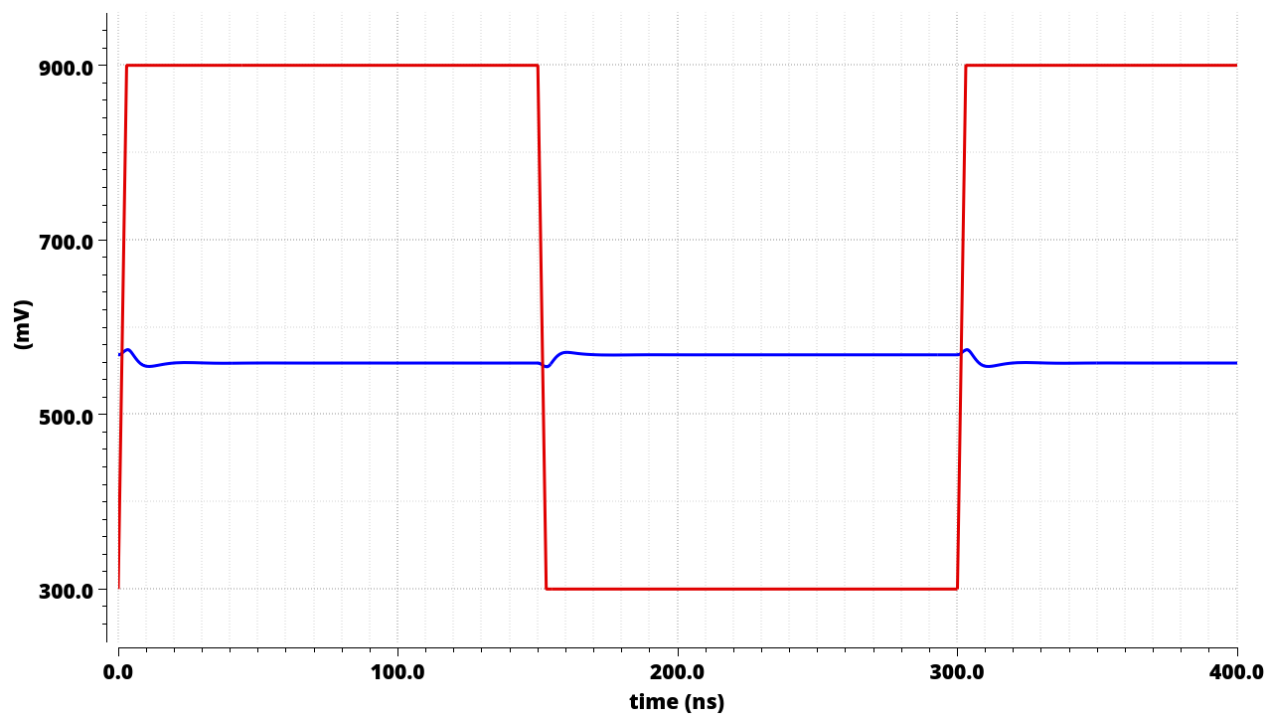


Figure 16: Step Response for Common Mode

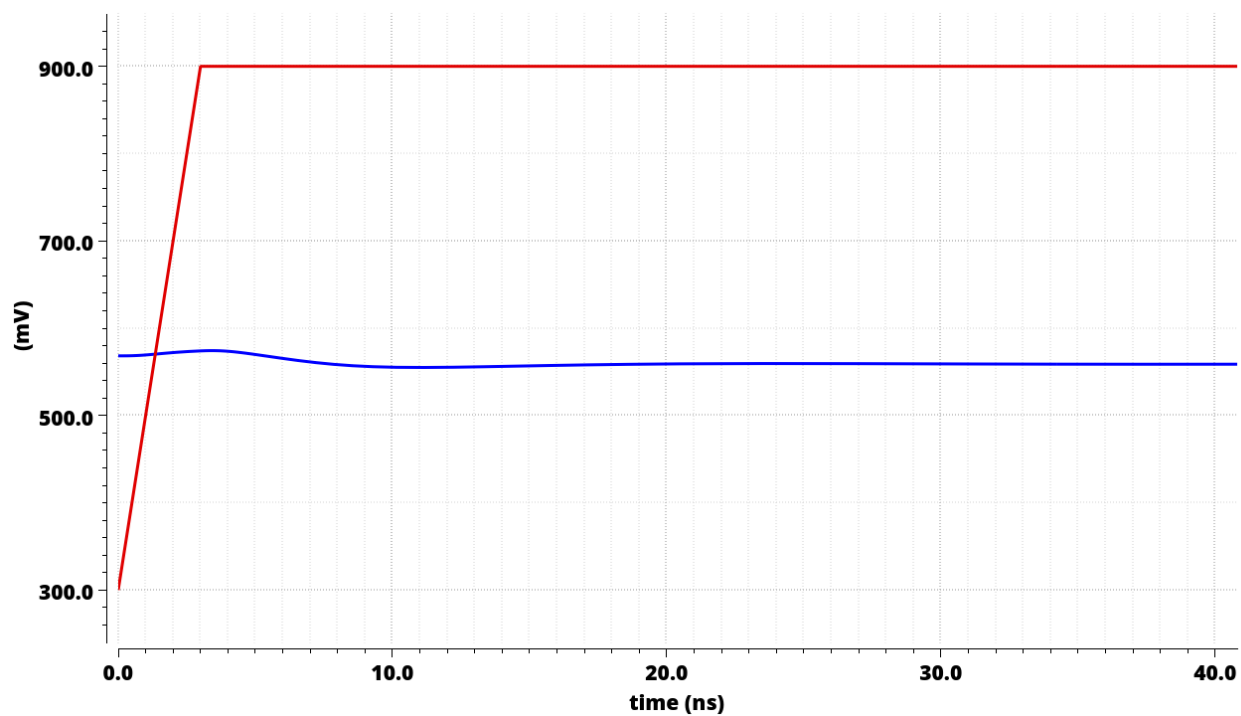


Figure 17: Step Response for Common Mode (zoomed to rising edge)

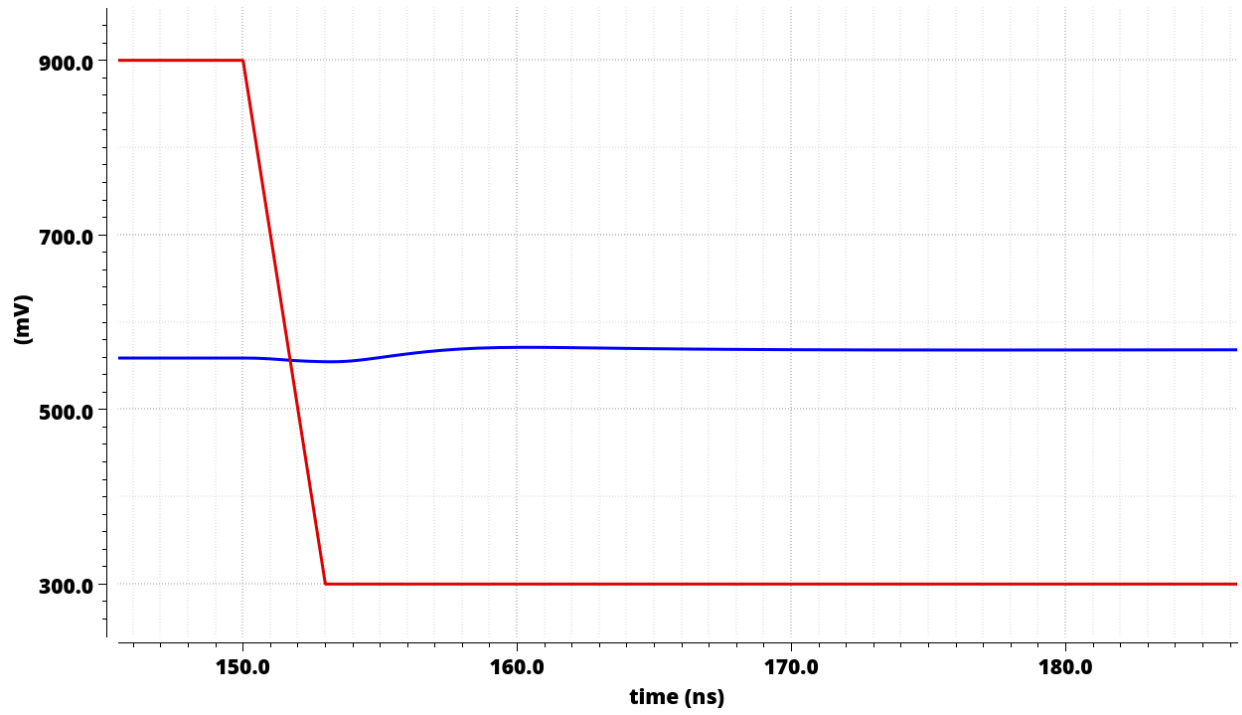


Figure 18: Step Response for Common Mode (zoomed to falling edge)

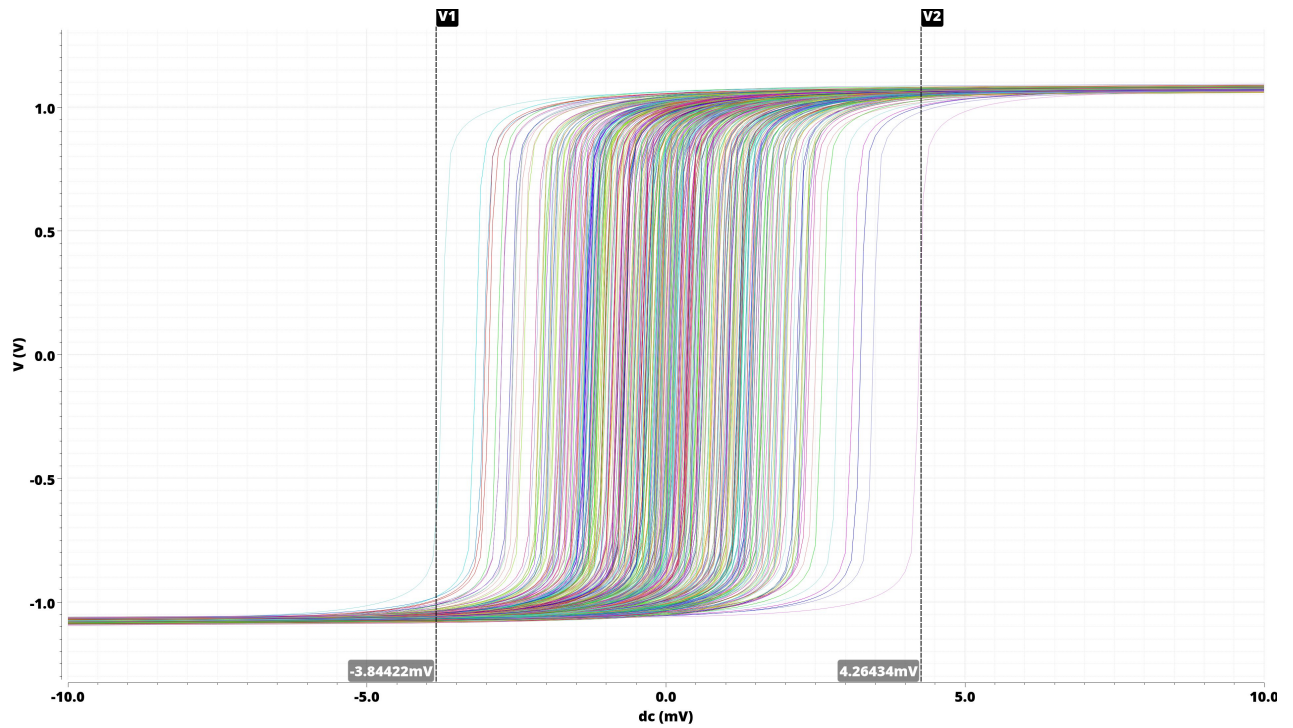


Figure 19: DC Differential Output vs Differential Input with 500 Samples Monte Carlo

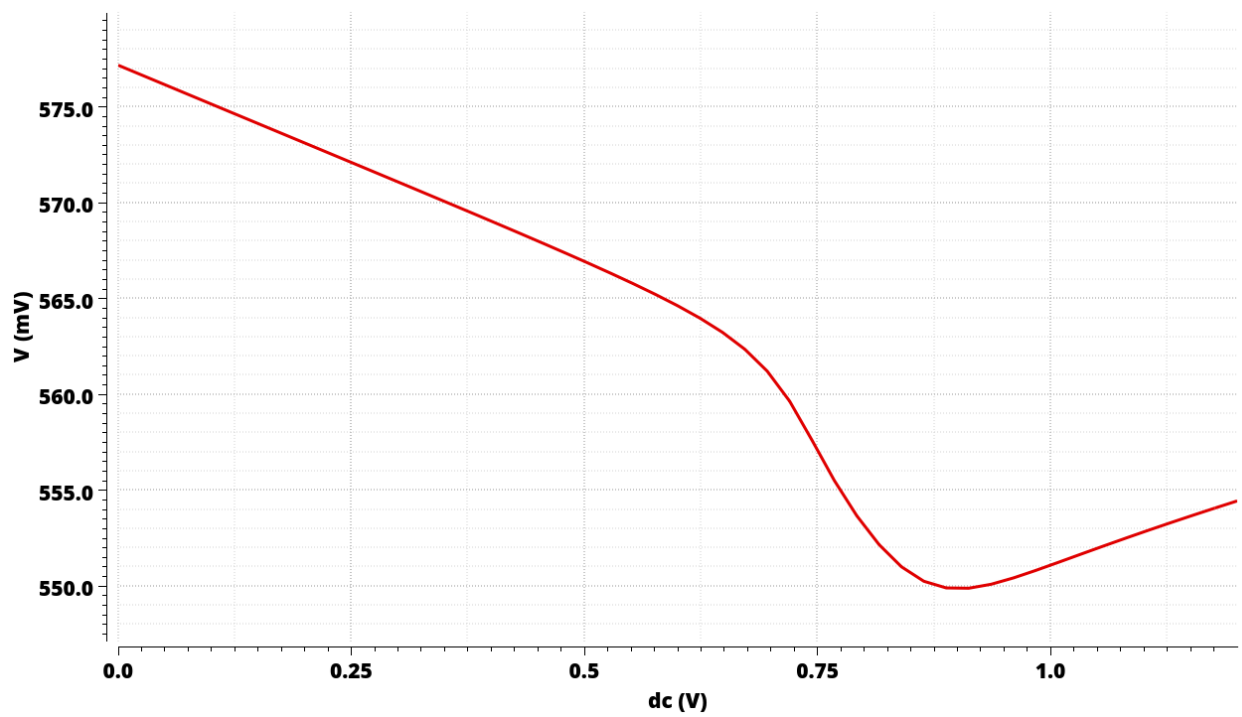


Figure 20: Output CM vs Input CM

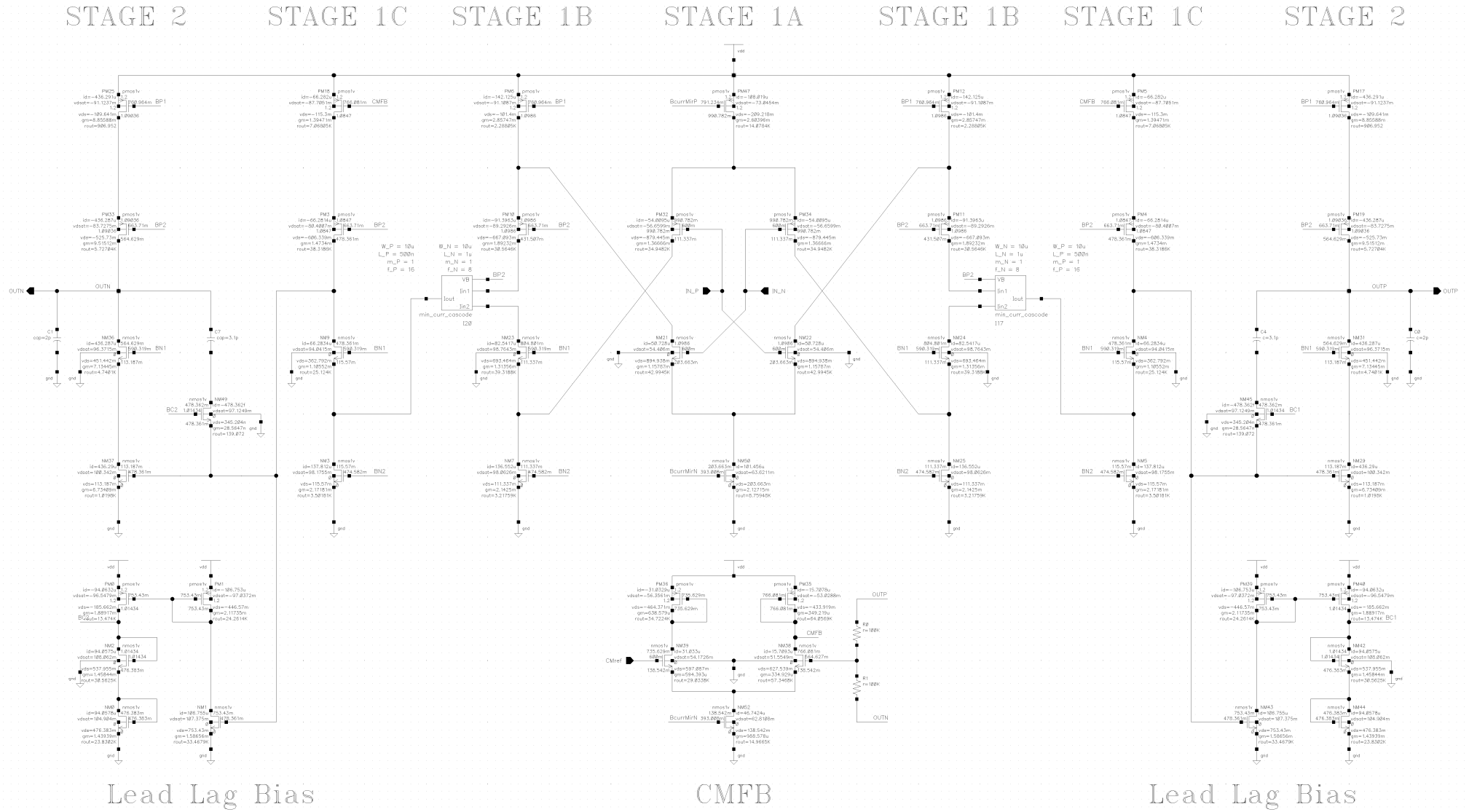


Figure 22: Final Schematic of OTA (DC Operating Point at Input CM = 0.6 V)

References

- [1] E. Sánchez-Sinencio, “Rail-to-rail op amps,” <https://picture.iczhiku.com/resource/eetop/whKfZWhwljWWgxNB.pdf>, [Accessed 12-03-2026].
- [2] C. Hwang, A. Motamed, and M. Ismail, “Universal constant-g/sub m/input-stage architectures for low-voltage op amps,” *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, vol. 42, no. 11, pp. 886–895, 1995.